

Lecture 17: Constructions by Straightedge and Compass

MAT205 Abstract Algebra II

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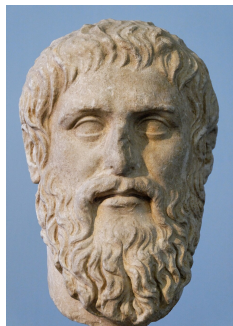
Part I: Constructible Numbers and the Field F

Doubling the Cube

The Delian altar (Plutarch, Eutocius). About 430 BC, plague on Delos; Apollo's oracle ordered the cubic altar doubled. The islanders doubled every edge—volume octupled, plague stayed. Plato told them the god wanted geometry, not arithmetic.

The problem. Given a segment of length 1, construct one of length $\sqrt[3]{2}$ by straightedge and compass.

Ancient workarounds. Hippocrates of Chios reduced this to finding two mean proportionals between 1 and 2; Menaechmus solved that with conics. Source: Eratosthenes' letter to Ptolemy III, lost itself but copied verbatim into Eutocius's commentary on Archimedes (c. 500 AD).



Plato (427–347 BC)
Athenian philosopher;
transmits the
Delian story.

Squaring the Circle

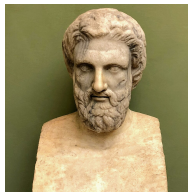
Earliest traces. Anaxagoras, jailed in Athens for impiety (he taught that the sun was a hot stone), is said by Plutarch to have worked on squaring the circle in prison—the first written trace. Aristophanes' *Birds* (414 BC) sends Meton on stage with a ruler, promising to “make the circle square”—already a household joke.

The problem. Given the unit disk, construct a square of equal area—equivalently, $\sqrt{\pi}$ —by straightedge and compass.

Hippocrates' lunes. Hippocrates of Chios squared certain crescents between two circular arcs—the first exact squaring of a curvilinear region. The method did not extend to the circle.



Anaxagoras (c. 500–428 BC)



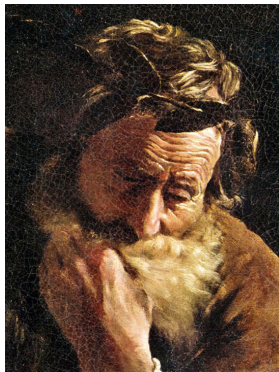
Aristophanes (c. 446–386 BC)

Trisecting the Angle

No founding myth. The problem arose naturally and stayed open. Hippias of Elis (c. 420 BC) invented the *quadratrix*, a curve that trisects any angle but is not itself constructible by line and circle. Archimedes added a clean *neusis* (marked-ruler) trisection—an extra rule, not a straightedge construction.

The problem. Given an angle θ , construct $\theta/3$ with straightedge and compass alone. Already 60° resists: trisecting it is the same as constructing $\cos 20^\circ$.

Classification (Pappus). The *Mathematical Collection* IV (c. 320 AD) classes such problems as “solid”—reachable with conics, suspected unreachable with line and circle. Wantzel (1837) proved the impossibility.

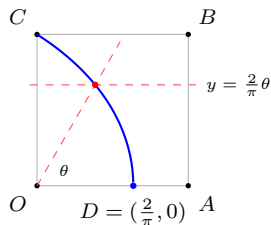


Archimedes (c. 287–212 BC)
Syracusan polymath;
exhaustion, π bounds;
neusis trisection.

The Quadratrix of Hippias

Definition. In the unit square $OABC$, rotate a radial line from OA to OC uniformly while a horizontal slides uniformly from OA to CB in equal time. Their intersection traces the *quadratrix*: at time t , angle $\theta = \frac{\pi}{2}t$, height $y = t$, so $y = \frac{2}{\pi}\theta$ along the curve.

Trisects, squares, but not Euclidean. Angle is proportional to height: divide the height by 3 (line and compass) to trisect θ . As $\theta \rightarrow 0$ the curve meets OA at $D = (\frac{2}{\pi}, 0)$; from $2/\pi$ one builds π , then $\sqrt{\pi}$ (mean proportional), squaring the unit disk. But $2/\pi$ is transcendental, so straightedge and compass cannot reach D : the curve needs two simultaneous uniform motions, which line and circle alone cannot produce.



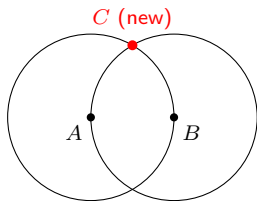
Constructed Points and Constructible Numbers

Constructed points are built recursively from $(0, 0)$ and $(1, 0)$:

- draw the *line* through two constructed points;
- draw the *circle* centered at a constructed point through a constructed point;
- a new point is an intersection of two such loci.

A real number α is **constructible** if some constructed segment has length $|\alpha|$; a point (x, y) is constructible iff x and y are.

Classical Euclidean moves are available: erect/drop perpendiculars, copy a length, draw parallels, bisect segments and angles. From the unit segment one locates every point with rational coordinates.



two unit circles centered at $A = (0, 0)$ and $B = (1, 0)$ meet at the new point

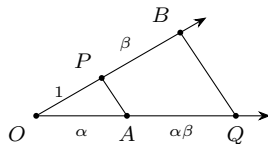
$$C = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right) \text{ (Euclid I.1)}$$

Constructible Numbers Form a Field

Theorem. If α, β are constructible, then so are $\alpha + \beta$, $\alpha - \beta$, $\alpha\beta$, and α/β (for $\beta \neq 0$).

Proof.

- *Sum/difference:* lay the length $|\beta|$ off along the line carrying $|\alpha|$ with the compass.
- *Product:* on one ray from O lay $|OP| = 1$ then $|PB| = |\beta|$; on another lay $|OA| = |\alpha|$. The parallel to PA through B meets ray OA at Q , and by the intercept theorem $\frac{|OP|}{|PB|} = \frac{|OA|}{|AQ|}$, so $|AQ| = |\alpha\beta|$.
- *Quotient:* the analogous parallel construction yields $|\alpha/\beta|$. ■



product by similar triangles: $PA \parallel BQ$

Corollary. $F := \{\text{constructible reals}\}$ is a subfield of $\mathbb{R}$, and $\mathbb{Q} \subseteq F$.

Square Roots Are Constructible

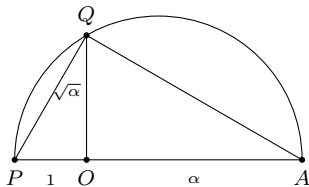
Theorem. If $\alpha > 0$ is constructible, then $\sqrt{\alpha}$ is constructible.

Proof (geometric mean). On a line mark P, O, A with $|PO| = 1$ and $|OA| = \alpha$. Draw the semicircle on diameter PA and the perpendicular to PA at O , meeting the semicircle at Q . By Thales' theorem the angle $\angle PQA$ is right, so triangles OPQ and OQA are similar:

$$\frac{|OQ|}{|OP|} = \frac{|OA|}{|OQ|} \implies |OQ|^2 = 1 \cdot \alpha = \alpha,$$

hence $|OQ| = \sqrt{\alpha}$. ■

So F is a field that is also closed under $\sqrt{}$ of positive elements — the one operation a compass adds beyond arithmetic.



Each Step Adjoins at Most One Square Root

Suppose every point built so far has coordinates in a subfield $H \subseteq \mathbb{R}$ (start $H = \mathbb{Q}$).
Then:

- lines through two such points: $ax + by + c = 0$ with $a, b, c \in H$;
- circles centered at such a point through such a point: $x^2 + y^2 + dx + ey + f = 0$ with $d, e, f \in H$.

A new point is an intersection of two of these:

- **line** $\cap$ **line**: two linear equations over H — the solution stays in H , no new number;
- **circle** $\cap$ **circle**: subtracting the equations cancels $x^2 + y^2$, leaving a line over H (the *radical axis*); this reduces to the next case;
- **line** $\cap$ **circle**: substitution gives a quadratic over H , whose roots lie in $H(\sqrt{d})$ for some $d \in H$, $d \geq 0$.

So each construction step keeps the coordinates inside H or a quadratic extension $H(\sqrt{d})$: the degree over the current field is 1 or 2.

The Tower Characterization of F

Theorem. For $\alpha \in \mathbb{R}$, the following are equivalent.

1. α is constructible.
2. There is a tower $\mathbb{Q} = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_n$ with $\alpha \in K_n$ and each $[K_i : K_{i-1}] = 2$ (so $K_i = K_{i-1}(\sqrt{d_i})$, $d_i \in K_{i-1}$, $d_i > 0$).
3. α is obtained from $\mathbb{Q}$ by finitely many field operations and square roots of positive numbers.

Proof.

- $(1) \Rightarrow (2)$: each new point adjoins one square root; collect them into a tower whose top field K_n contains α .
- $(2) \Rightarrow (3)$: each step $K_{i-1}(\sqrt{d_i})$ is field operations plus one square root.
- $(3) \Rightarrow (1)$: F is a field containing $\mathbb{Q}$ and closed under positive square roots, so it contains every such expression. ■

Part II: The Degree Criterion

Constructible Numbers Have 2-Power Degree

Corollary. If $\alpha \in \mathbb{R}$ is constructible, then $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 2^r$ for some integer $r \geq 0$.

Proof. By the tower characterization $\alpha \in K_n$ with each $[K_i : K_{i-1}] = 2$, so by the tower law

$$[K_n : \mathbb{Q}] = \prod_{i=1}^n [K_i : K_{i-1}] = 2^n.$$

Since $\mathbb{Q} \subseteq \mathbb{Q}(\alpha) \subseteq K_n$, the tower law gives $[K_n : \mathbb{Q}] = [K_n : \mathbb{Q}(\alpha)] [\mathbb{Q}(\alpha) : \mathbb{Q}]$, so $[\mathbb{Q}(\alpha) : \mathbb{Q}]$ divides 2^n and is a power of 2. ■

This criterion is due to Pierre Wantzel (1837). **Use:** to show a length is *not* constructible, exhibit a number whose degree over $\mathbb{Q}$ is not a power of 2 — most cheaply, degree 3.

Necessary, Not Sufficient

The converse of the degree criterion is **false**.

There exist α with $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 4$ that are not constructible: a root of an irreducible quartic whose splitting field has Galois group S_4 has degree 4, yet cannot be reached by a tower of quadratic extensions.

The sharp condition is Galois-theoretic: α is constructible exactly when the *Galois closure* of $\mathbb{Q}(\alpha)/\mathbb{Q}$ — the smallest Galois extension of $\mathbb{Q}$ containing α — is a 2-group. (Over $\mathbb{Q}$ every algebraic extension is separable, so the Galois closure coincides with the normal closure of L15 and equals the splitting field of $\text{irr}(\alpha, \mathbb{Q})$.) We prove this once the Galois correspondence is in hand.

For the three classical problems the necessary direction already suffices, so we settle them first.

Part III: The Three Impossibility Proofs

Doubling the Cube Is Impossible

Theorem. One cannot construct, with straightedge and compass, the edge of a cube of twice the volume of a given cube.

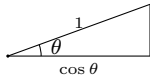
Proof. Take the given edge as the unit; the target edge has length $\sqrt[3]{2}$. The polynomial $x^3 - 2$ is irreducible over $\mathbb{Q}$ (Eisenstein at $p = 2$), so

$$[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3.$$

Since 3 is not a power of 2, the degree criterion shows $\sqrt[3]{2}$ is not constructible. ■

Trisecting the Angle: Reducing to a Cubic

Lemma (angle $\leftrightarrow$ cosine). An angle θ is constructible iff a segment of length $|\cos \theta|$ is constructible. (*With the unit segment as hypotenuse, the adjacent leg has length $|\cos \theta|$.*)



Triple-angle identity. From $\cos 3\theta = \cos(2\theta + \theta)$ and the addition formulas,

$$\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta.$$

Put $\theta = 20^\circ$, so $\cos 3\theta = \cos 60^\circ = \frac{1}{2}$. With $\alpha = \cos 20^\circ$,

$$4\alpha^3 - 3\alpha = \frac{1}{2}, \quad \text{i.e.} \quad 8\alpha^3 - 6\alpha - 1 = 0.$$

Trisecting the Angle Is Impossible

Theorem. The angle 60° cannot be trisected by straightedge and compass.

Proof. 60° is constructible (equilateral triangle). Trisecting it means constructing $\alpha = \cos 20^\circ$, a root of $f(x) = 8x^3 - 6x - 1$. Substituting $x = y + \frac{1}{2}$,

$$f\left(y + \frac{1}{2}\right) = 8y^3 + 12y^2 - 3,$$

which is Eisenstein at $p = 3$ ($3 \nmid 8$, $3 \mid 12$, $3 \mid -3$, $9 \nmid -3$), hence irreducible over $\mathbb{Q}$. So f is irreducible too, and

$$[\mathbb{Q}(\cos 20^\circ) : \mathbb{Q}] = 3.$$

As 3 is not a power of 2, $\cos 20^\circ$ is not constructible, so 60° is not trisectable. ■

Some angles do trisect (e.g. $90^\circ \rightarrow 30^\circ$); the theorem is that no single method trisects every angle.

Squaring the Circle Is Impossible

Theorem. One cannot construct a square equal in area to a given circle.

Cited input (Lindemann, 1882). π is *transcendental* over $\mathbb{Q}$: it is a root of no nonzero polynomial in $\mathbb{Q}[x]$. (The proof is beyond this course; we use it as a black box.)

Proof. For the unit circle (area π) the square has side $\sqrt{\pi}$. If $\sqrt{\pi}$ were constructible then so would $\pi = (\sqrt{\pi})^2$ be, hence π would be algebraic of degree a power of 2. But a transcendental number lies in no finite extension of $\mathbb{Q}$ — contradiction. ■



Ferdinand von Lindemann
(1852–1939): proved π
transcendental in 1882.

Part IV: Regular Polygons and the Galois Criterion

Constructibility in the Plane

Identify the plane with $\mathbb{C}$. A complex number $z = a + bi$ is **constructible** iff $a, b \in F$; the constructible complex numbers form a subfield $\mathbb{K} \subseteq \mathbb{C}$ containing $\mathbb{Q}$.

Closed under square roots. If $z = \rho e^{i\psi}$ is constructible, one constructs $\sqrt{\rho}$ (geometric mean) and bisects the angle ψ , giving $\sqrt{\rho} e^{i\psi/2}$. So $\mathbb{K}$ is closed under $\sqrt{}$.

Complex tower characterization. For $z \in \mathbb{C}$, z is constructible iff z lies in a tower $\mathbb{Q} = K_0 \subseteq \cdots \subseteq K_n$ of subfields of $\mathbb{C}$ with each $[K_i : K_{i-1}] = 2$. (Each step adjoins a $\sqrt{}$, which $\mathbb{K}$ contains; conversely $z = a + bi$ with $a, b \in F$ sits in a real quadratic tower, and $i = \sqrt{-1}$ is one more quadratic step.)

The Regular n -gon Reduces to $\cos(2\pi/n)$

Inscribe the regular n -gon in the unit circle with one vertex at 1. Its vertices are the powers ζ_n^k of

$$\zeta_n = e^{2\pi i/n} = \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}.$$

Therefore:

- the n -gon is constructible $\iff \zeta_n$ is constructible;
- ζ_n is constructible $\iff \cos(2\pi/n)$ is constructible, since then $\sin(2\pi/n) = \sqrt{1 - \cos^2(2\pi/n)}$ is constructible too.

So the geometric question becomes: *for which n is $\cos(2\pi/n)$ constructible?* The Galois criterion answers it.

The Galois Criterion for Constructibility

Theorem. Let $\alpha \in \mathbb{C}$ be algebraic over $\mathbb{Q}$, and let L be the Galois closure of $\mathbb{Q}(\alpha)/\mathbb{Q}$ (the splitting field of $\text{irr}(\alpha, \mathbb{Q})$). Then

$$\alpha \text{ constructible} \iff [L : \mathbb{Q}] = 2^m \iff \text{Gal}(L/\mathbb{Q}) \text{ is a 2-group.}$$

Proof of $\Leftarrow$ (sufficiency). Let $G = \text{Gal}(L/\mathbb{Q})$ have order 2^m . A finite 2-group has a chain of subgroups

$$G = G_0 \triangleright G_1 \triangleright \cdots \triangleright G_m = \{1\}, \quad [G_i : G_{i+1}] = 2.$$

By the Galois correspondence this gives a tower of fields

$$\mathbb{Q} = F_0 \subseteq F_1 \subseteq \cdots \subseteq F_m = L, \quad [F_{i+1} : F_i] = [G_i : G_{i+1}] = 2.$$

Each step is a quadratic extension, so every element of L — in particular α — lies in a quadratic tower and is constructible.

The Galois Criterion: Necessity

Proof of $\Rightarrow$. If α is constructible it lies in a quadratic tower $\mathbb{Q} = K_0 \subseteq \cdots \subseteq K_n$. For each $\mathbb{Q}$ -embedding $\sigma : K_n \hookrightarrow \mathbb{C}$, the image σK_n is again a quadratic tower $(\sigma K_i = \sigma K_{i-1}(\sqrt{\sigma d_i}))$.

Let M be the compositum of all these conjugate towers. Adjoining the finitely many square roots $\sqrt{\sigma d_i}$ one at a time builds M by steps of degree 1 or 2, so $[M : \mathbb{Q}] = 2^k$. Now M contains every conjugate $\sigma\alpha$, hence contains the splitting field $L \subseteq M$. By the tower law $[L : \mathbb{Q}]$ divides 2^k , so it is a power of 2; and as $L/\mathbb{Q}$ is Galois, $|\text{Gal}(L/\mathbb{Q})| = [L : \mathbb{Q}] = 2^m$. ■

This recovers the degree criterion ($[\mathbb{Q}(\alpha) : \mathbb{Q}]$ divides 2^m) and explains its failure as a converse: the *closure* must be a 2-group, not merely $\mathbb{Q}(\alpha)$ of 2-power degree.

Cyclotomic Degree and Galois Group

The n -gon field. The minimal polynomial of ζ_n over $\mathbb{Q}$ is the cyclotomic polynomial Φ_n , irreducible of degree $\varphi(n)$ (Euler's totient). Hence

$$[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n), \quad \text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times,$$

via $\sigma_a : \zeta_n \mapsto \zeta_n^a$ for $\gcd(a, n) = 1$. (The irreducibility of Φ_n is a classical theorem of Gauss; the group is abelian of order $\varphi(n)$.)

Consequence. $\mathbb{Q}(\zeta_n)/\mathbb{Q}$ is Galois and abelian, so it is its own Galois closure. By the Galois criterion,

$$\zeta_n \text{ constructible} \iff \varphi(n) \text{ is a power of } 2.$$

The Gauss–Wantzel Theorem

Fermat prime. A prime of the form $F_t = 2^{2^t} + 1$; the known ones are 3, 5, 17, 257, 65537.

Theorem. The regular n -gon ($n \geq 3$) is constructible iff

$$n = 2^k p_1 p_2 \cdots p_s,$$

with $k \geq 0$ and the p_i distinct Fermat primes.

Proof. The n -gon is constructible iff $\varphi(n)$ is a power of 2. For $n = 2^{a_0} p_1^{a_1} \cdots p_s^{a_s}$,

$$\varphi(n) = \varphi(2^{a_0}) \prod_i p_i^{a_i-1} (p_i - 1),$$

and $\varphi(2^{a_0})$ is always a power of 2. So $\varphi(n)$ is a power of 2 iff every $a_i = 1$ and every $p_i - 1$ is a power of 2; and $p - 1 = 2^e$ forces $e = 2^t$ (an odd factor $v > 1$ of $e = uv$ would give $2^u + 1 \mid 2^e + 1 = p$), i.e. p a Fermat prime. ■

Constructible: 3, 4, 5, 6, 8, 10, 12, 15, 16, 17, ... Not: 7, 9, 11, 13, 14, ... ($\varphi(7) = \varphi(9) = 6$).

Gauss and the Regular 17-gon

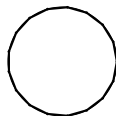
In 1796, aged 19, Gauss found the regular 17-**gon** constructible — $\varphi(17) = 16 = 2^4$ — and opened his *Disquisitiones Arithmeticae* (1801) with it. He gave the explicit value

$$\begin{aligned}\cos \frac{2\pi}{17} = & -\frac{1}{16} + \frac{1}{16}\sqrt{17} + \frac{1}{16}\sqrt{34 - 2\sqrt{17}} \\ & + \frac{1}{8}\sqrt{17 + 3\sqrt{17} - \sqrt{34 - 2\sqrt{17}} - 2\sqrt{34 + 2\sqrt{17}}},\end{aligned}$$

a tower of nested square roots — exactly a quadratic tower. Gauss stated the full classification and proved the constructible half.



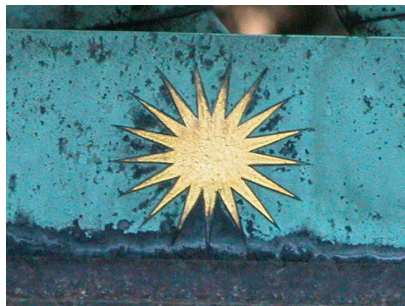
C. F. Gauss (1777–1855).



Gauss's Tomb and Memorial



Gauss's grave, Albani-Friedhof, Göttingen.



17-pointed star on the Gauss-Denkmal
(Braunschweig; statue 1880).

Gauss reportedly asked for a regular 17-gon on his tombstone; the mason refused, saying it would look like a circle. The 17-pointed star on the Brunswick memorial stands in.

The Five Squarable Lunes

Hippocrates' partial success (c. 440 BC). Certain *lunes*—crescents bounded by two circular arcs—can be squared exactly with straightedge and compass; this is the first such squaring of any curvilinear region. Hippocrates of Chios found three, with angular arc-ratios

$$2:1, \quad 3:1, \quad 3:2.$$

Wallenius (1766). Two more, with ratios 5:1 and 5:3.

Tschebotaröw (1934) – Dorodnov (1947). Among all lunes whose outer and inner arcs subtend angles in a rational ratio, the five above are the *only* ones squarable by straightedge and compass. The classification is complete.

The circle itself is no lune (only one arc, no second to set in ratio). Lindemann (1882) settled it separately.

Mohr–Mascheroni and Poncelet–Steiner

Two equivalence theorems. Neither enlarges the constructible set, but each shows one of the two classical instruments is redundant.

Mohr (1672) – Mascheroni (1797). Every straightedge-and-compass construction can be carried out by the *compass alone*—if a "line through two given points" is identified with the pair of points (no physical line drawn). Mascheroni published this in *La geometria del compasso* (1797); Georg Mohr's earlier *Euclides Danicus* (1672) was forgotten until 1928.

Poncelet (1822) – Steiner (1833). Every straightedge-and-compass construction can be carried out by the *straightedge alone*, given one auxiliary circle in the plane with its centre marked. Conjectured by Poncelet, proved by Jakob Steiner. The marked centre is essential: Steiner also showed an unmarked circle is not enough.

Reading. The genuinely new operation in Euclidean construction is the compass; the straightedge is dispensable in one direction, and almost dispensable in the other.